

Table S1: LLM Models and Their Capabilities

Model	Token Limit	Context Handling	Task Specialization	Additional Features
GPT-4oV	128,000 tokens	Moderate contexts	Pathway analysis and moderate text extraction	Vision capabilities, faster processing, cost-efficient
Claude-3.5V	200,000 tokens	Extensive data	Large textual data processing	Enhanced coding abilities, extended context retention
Gemini-1.5V	Millions of tokens	Long-context tasks	Massive dataset processing	Multimodal input, native tool integration, scalable context
Llama 3.2V	128,000 tokens	Complex interactions	Complex biological interaction analysis	Vision-language integration, on-device deployment options

Table S2. GPV Prompts for Gene Interaction Extraction from Pathway Diagrams.

1	2
As a Molecular Pathway Specialist, Extract and classify gene interactions from the pathway diagram with precision. The diagram uses T-bars (---) to indicate "Inhibition," with dashed T-bars for "Indirect Inhibition," and arrows (→) to show "Activation," with dashed arrows representing "Indirect Activation." The arrows and T-bars indicate the direction from gene1 to gene2. Carefully extract each interaction, classify it as "direct" or "indirect," and format it as "gene1 relationship gene2." Provide a comprehensive list of gene interactions identified from the pathway diagram.	As a Gene Interaction Expert, Extract and categorize gene interactions from the pathway diagram with precision. In the pathway diagram, "Inhibition" is indicated by T-bars (---), with dashed T-bars for "Indirect Inhibition," and "Activation" is shown by arrows (→), with dashed arrows for "Indirect Activation." The direction of arrows and T-bars shows the relationship from gene1 to gene2. Identify and classify each interaction, distinguishing between "direct" and "indirect," and format them as "gene1 relationship gene2." Provide a detailed list of gene interactions identified from the pathway diagram.
3	4
As a Gene Pathway Specialist, accurately extract and classify gene relationships from the provided pathway diagram. Symbols in the diagram represent interactions: T-bars (---) for "Inhibition," with dashed T-bars indicating "Indirect Inhibition," and arrows (→) for "Activation," with dashed arrows representing "Indirect Activation." Arrows and T-bars indicate the direction from gene1 to gene2. Extract and differentiate each interaction, classifying it as "direct" or "indirect," and format them as "gene1 relationship gene2." Submit a comprehensive list of gene interactions identified from the pathway diagram.	As a Gene Network Specialist, accurately extract and document gene interactions from the provided pathway diagram. The diagram uses T-bars (---) to indicate "Inhibition," with dashed T-bars for "Indirect Inhibition," and arrows (→) to show "Activation," with dashed arrows representing "Indirect Activation." The arrows and T-bars indicate the direction from gene1 to gene2. Identify and categorize each interaction as "direct" or "indirect," and format them as "gene1 relationship gene2." Provide a comprehensive list of gene interactions identified from the pathway diagram.
5	
As a Pathway Interaction Analyst, accurately extract gene interactions from the provided pathway diagram. The diagram uses T-bars (---) to indicate "Inhibition," with dashed T-bars for "Indirect Inhibition," and arrows (→) to show "Activation," with dashed arrows representing "Indirect Activation." The arrows and T-bars indicate the direction from gene1 to gene2. Identify each interaction, distinguish between "direct" and "indirect," and format them as "gene1 relationship gene2." Provide a comprehensive list of gene interactions identified from the pathway diagram.	

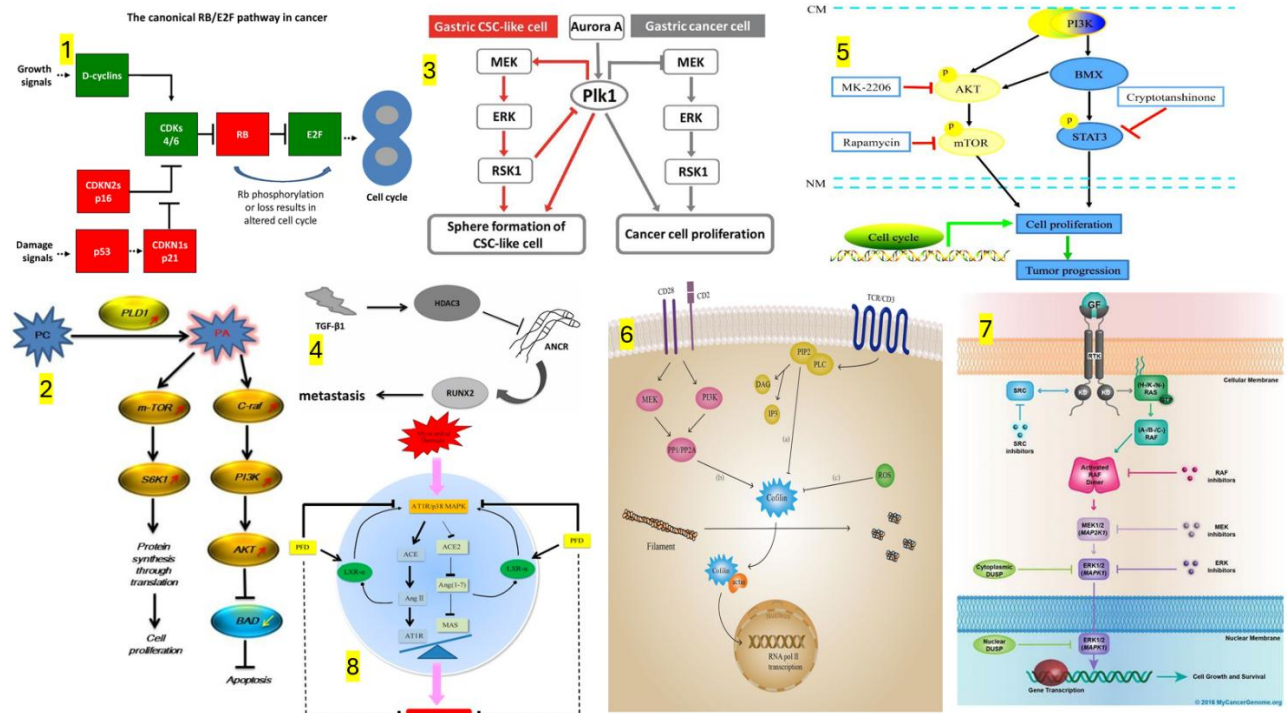


Figure S1: Evaluation of different prompts across various pathway diagrams.

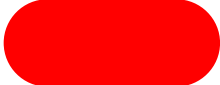
	Correct predictions	The model correctly identifies a gene relationship as it exists in the data. Example: Gene A activates Gene B, and the model predicts "activation."
	Incorrect predictions (There are two types of incorrect predictions)	Wrong Relationship Type: ✓ Example: Gene A activates Gene B, but the model predicts "inhibition" instead. ✓ Error: The model detects the relationship but misclassifies its nature. Non-Existent Relationship: ✓ Example: The model predicts that Gene A interacts with Gene B, but no such relationship exists in the ground truth. ✓ Error: The model fabricates a connection that is not present in the actual data.
	Missing predictions	A real gene relationship exists in the data, but the model completely misses it and does not predict anything. Example: Gene A activates Gene B, but the model does not recognize any connection between them.

Table S3. Results for each prompt's performance across pathway images

Image-1

Image 1 Prompt -1		T1	
GPT-4oV	Claude-3.5V	Llama-3.2V	Gemini-1.5V
Growth signals Activation D-cyclins D-cyclins Activation CDKs 4/6 CDKs 4/6 Inhibition RB RB Inhibition E2F Damage signals Activation p53 p53 Activation CDKN1s (p21) Damage signals Activation CDKN1s CDKN1s (p21) Inhibition CDKs 4/6 CDKN2s (p16) Inhibition CDKs 4/6	Growth signals Activation D-cyclins D-cyclins Activation CDKs 4/6 CDKs 4/6 Inhibition RB RB Inhibition E2F Damage signals Activation p53 p53 Activation CDKN1s (p21) Damage signals Activation CDKN1s CDKN1s (p21) Inhibition CDKs 4/6 CDKN2s (p16) Inhibition CDKs 4/6	Growth signals Activation D-cyclins D-cyclins Activation CDKs 4/6 CDKs 4/6 Inhibition RB RB Inhibition E2F Damage signals Activation p53 p53 Activation CDKN1s (p21) CDKN1s (p21) Inhibition CDKs 4/6 CDKN2s (p16) Inhibition CDKs 4/6	Growth signals Activation D-cyclins D-cyclins Activation CDKs 4/6 CDKs 4/6 Inhibition RB RB Inhibition E2F Damage signals Activation p53 p53 Activation CDKN1s (p21) CDKN1s (p21) Inhibition CDKs 4/6 CDKN2s (p16) Activation CDKs 4/6
T2			
GPT-4oV	Claude-3.5V	Llama-3.2V	Gemini-1.5V
Growth signals Activation D-cyclins D-cyclins Activation CDKs 4/6 CDKs 4/6 Inhibition RB RB Inhibition E2F Damage signals Activation p53 p53 Activation CDKN1s (p21) Damage signals Activation CDKN1s (p21) CDKN1s (p21) Inhibition CDKs 4/6 CDKN2s (p16) Inhibition CDKs 4/6	Growth signals Activation D-cyclins D-cyclins Activation CDKs 4/6 CDKs 4/6 Inhibition RB RB Inhibition E2F Damage signals Activation p53 p53 Activation CDKN1s (p21) Damage signals Activation CDKN1s (p21) CDKN1s (p21) Inhibition CDKs 4/6 CDKN2s (p16) Inhibition CDKs 4/6	Growth signals Activation D-cyclins D-cyclins Activation CDKs 4/6 CDKs 4/6 Inhibition RB RB Inhibition E2F Damage signals Activation p53 p53 Activation CDKN1s (p21) CDKN1s (p21) Inhibition CDKs 4/6 CDKN2s (p16) Inhibition CDKs 4/6	Growth signals Activation D-cyclins D-cyclins Activation CDKs 4/6 CDKs 4/6 Inhibition RB RB Inhibition E2F Damage signals Activation p53 p53 Activation CDKN1s (p21) CDKN1s (p21) Inhibition CDKs 4/6 CDKN2s (p16) Activation CDKs 4/6
Image 1 - Prompt (2)		T1	
GPT-4oV	Claude-3.5V	Llama-3.2V	Gemini-1.5V
D-cyclins Activation CDKs 4/6. CDKs 4/6 Inhibition RB. RB Inhibition E2F. CDKN2s Inhibition CDKs 4/6. p53 Activation CDKN1s (p21). CDKN1s (p21) Inhibition CDKs 4/6.	D-cyclins Activation CDKs 4/6. CDKs 4/6 Activation RB. RB Inhibition E2F. CDKN2s Inhibition CDKs 4/6. p53 Activation CDKN1s (p21). CDKN1s (p21) Inhibition CDKs 4/6.	D-cyclins Activation CDKs 4/6. CDKs 4/6 Inhibition RB. RB Inhibition E2F. CDKN2s Inhibition CDKs 4/6. p53 Activation CDKN1s (p21). CDKN1s (p21) Inhibition CDKs 4/6.	D-cyclins Activation CDKs 4/6. CDKs 4/6 Inhibition RB. RB Inhibition E2F. CDKN2s Activation CDKs 4/6. p53 Activation CDKN1s (p21). CDKN1s (p21) Inhibition CDKs 4/6.
T2			
GPT-4oV	Claude-3.5V	Llama-3.2V	Gemini-1.5V
D-cyclins Activation CDKs 4/6. CDKs 4/6 Inhibition RB.	D-cyclins Activation CDKs 4/6. CDKs 4/6 Activation RB.	D-cyclins Activation CDKs 4/6. CDKs 4/6 Inhibition RB.	D-cyclins Activation CDKs 4/6. CDKs 4/6 Inhibition RB.

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[illegible]

Image-5

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Cryptotansh inhibition STAT3	Cryptotanshi inhibition STAT3	Cryptotans inhibition STAT3	Cryptotanshinone inhibition STAT3
T2			
GPT-4oV	Claude-3.5V	Llama-3.2V	Gemini-1.5V
PI3K Activation BMX PI3K Activation AKT BMX Activation STAT3 BMX Activation AKT mTOR Activation Cell Proliferati Cell Prolife Activation Tumor Cell Cycle Activation Cell Prolife MK-2206 inhibition AKT Rapamycin Activation mTOR Cryptotansh inhibition STAT3	PI3K Activation BMX PI3K Activation AKT BMX Activation STAT3 BMX Activation AKT AKT Activation mTOR mTOR inhibition Cell Proliferation Cell Proliferation Activation Tumor Cell Cycle Activation Cell Prolife MK-2206 inhibition AKT Rapamycin inhibition mTOR Cryptotanshi inhibition STAT3	PI3K Activation BMX PI3K Activation AKT BMX Activation STAT3 BMX Activation AKT AKT Activation mTOR mTOR Activation Cell Prolife Cell Prolif Activation Tumor Cell Cycle Activation Cell Prolife MK-2206 inhibition AKT Rapamycin inhibition mTOR Cryptotans inhibition STAT3	PI3K Activation BMX PI3K Activation AKT BMX Activation STAT3 BMX Activation AKT AKT Activation mTOR mTOR inhibition Cell Proliferation Cell Proliferation Activation Tumor Cell Cycle inhibition Cell MK-2206 inhibition AKT Rapamycin inhibition mTOR Cryptotanshinone inhibition STAT3

Image-6

Image 6 - Prompt (1)			
T1			
GPT-4oV	Claude-3.5V	Llama-3.2V	Gemini-1.5V
CD28 Activation MEK CD28 Activation PI3K MEK Activation PP1/PP2A PI3K Activation PP1/PP2A PIP2 Activation PLC PLC Activation IP3 PLC Activation DAG IP3 activation Cofilin (a) PP1/PP2A Activation Cofilin (b) ROS inhibition Cofilin (c) Cofilin Activation Actin Actin Activation RNA pol II TCR/CD3 Activation PLC	CD28 Activation MEK CD28 Activation PI3K MEK Activation PP1/PP2A PI3K Activation PP1/PP2A PIP2 Activation PLC PLC Activation IP3 PLC Activation DAG IP3 activation Cofilin (a) PP1/PP2A Activation Cofilin (b) ROS inhibition Cofilin (c) Cofilin Activation Actin Actin Activation RNA pol II TCR/CD3 Activation PLC	CD28 Activation MEK CD28 Activation PI3K MEK Activation PP1/PP2A PI3K Activation PP1/PP2A PIP2 Activation PLC PIP2 Activation PLC PLC Activation IP3 PLC Activation DAG IP3 activation Cofilin (a) PP1/PP2A Activation Cofilin (b) ROS inhibition Cofilin (c) Cofilin Activation Actin Actin Activation RNA pol II TCR/CD3 Activation PLC	CD28 Activation PI3K MEK Activation PP1/PP2A PI3K Activation PP1/PP2A PIP2 Activation PLC PLC Activation IP3 PLC Activation DAG IP3 activation Cofilin (a) PP1/PP2A Activation Cofilin (b) ROS inhibition Cofilin (c) Cofilin Activation Actin Actin Activation RNA pol II TCR/CD3 Activation PLC
T2			
GPT-4oV	Claude-3.5V	Llama-3.2V	Gemini-1.5V
CD28 Activation MEK CD28 Activation PI3K MEK Activation PP1/PP2A PI3K Activation PP1/PP2A PIP2 Activation PLC PLC Activation IP3 PLC Activation DAG IP3 activation Cofilin (a) PP1/PP2A Activation Cofilin (b) ROS inhibition Cofilin (c) Cofilin Activation Actin Actin Activation RNA pol II TCR/CD3 Activation PLC	CD28 Activation MEK CD28 Activation PI3K MEK Activation PP1/PP2A PI3K Activation PP1/PP2A PIP2 Activation PLC PLC Activation IP3 PLC Activation DAG PP1/PP2A Activation Cofilin (b) ROS inhibition Cofilin (c) Cofilin Activation Actin Actin Activation RNA pol II TCR/CD3 Activation PLC	CD28 Activation MEK CD28 Activation PI3K MEK Activation PP1/PP2A PI3K Activation PP1/PP2A PIP2 Activation PLC PIP2 Activation PLC PLC Activation IP3 PLC Activation DAG IP3 activation Cofilin (a) PP1/PP2A Activation Cofilin (b) ROS inhibition Cofilin (c) Cofilin Activation Actin Actin Activation RNA pol II TCR/CD3 Activation PLC	CD28 Activation PI3K MEK Activation PP1/PP2A PI3K Activation PP1/PP2A PIP2 Activation PLC PLC Activation IP3 PLC Activation DAG PP1/PP2A Activation Cofilin (b) ROS inhibition Cofilin (c) Cofilin Activation Actin Actin Activation RNA pol II TCR/CD3 Activation PLC
Image 6 - Prompt (2)			
T1			
GPT-4oV	Claude-3.5V	Llama-3.2V	Gemini-1.5V
CD28 activation PI3K. CD28 Activation MEK. PI3K Activation PP1/PP2A MEK Activation PP1/PP2A PP1/PP2A Activation Cofilin, MEK Activation Cofilin,	CD28 activation PI3K. CD28 Activation MEK. PI3K Activation PP1/PP2A MEK Activation PP1/PP2A PP1/PP2A Activation Cofilin, MEK Activation Cofilin,	CD28 activation PI3K. CD28 Activation MEK. PI3K Activation PP1/PP2A MEK Activation PP1/PP2A PP1/PP2A Activation Cofilin, MEK Activation Cofilin,	CD28 activation PI3K. CD28 Activation MEK. PI3K Activation PP1/PP2A MEK Inhibition PP1/PP2A PP1/PP2A Activation Cofilin, PLC Activation DAG/IP3

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Image-7

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